



SCEGGS Darlinghurst

2010

HSC Assessment 2

11th June, 2010

Mathematics Extension 1

Outcomes Assessed: PE2, PE3, HE4, HE6 and HE7

General Instructions

- Time allowed – 70 minutes
- This paper has **four** questions
- Attempt **all** questions
- Answer all questions on the pad paper provided
- Begin each question on a **new page**
- Write your Student Number at the top of each page
- Mathematical templates, geometrical equipment and approved scientific calculators may be used
- A table of standard integrals is provided

Question	Calculus	Communication	Reasoning	Marks
1	/2	/3		/12
2	/2	/3	/2	/12
3	/5		/6	/12
4	/5	/1	/4	/12
TOTAL	/15	/7	/12	/48

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Total marks – 48**Attempt Questions 1–4**

Answer each question on the pad paper provided.
Write your student number at the top of each page.
Begin each question on a NEW page.

Marks**Question 1 (12 marks)**

- (a) If α , β , and γ are the roots of $P(x) = 2x^3 - x^2 - 8x + 4$, find 5
- (i) $\alpha + \beta + \gamma$
- (ii) $\alpha\beta + \beta\gamma + \alpha\gamma$
- (iii) $\alpha\beta\gamma$
- (iv) $\alpha^2 + \beta^2 + \gamma^2$
- (b) (i) State the domain and range of the function $f(x) = 3\cos^{-1} 2x$. 2
- (ii) Draw a neat sketch of the function $f(x) = 3\cos^{-1} 2x$, clearly labeling important features. 1
- (c) Find the exact value of $\int_0^4 \frac{3}{\sqrt{16-x^2}} dx$. 2
- (d) A function is given by the rule $f(x) = \frac{x+1}{x+2}$. 2
- Find the rule for the inverse function $f^{-1}(x)$.

End of Question 1

Question 2 (12 marks) Begin a NEW page.

- (a) (i) Sketch $y = 3 \sin x$ and $y = x$ for $0 \leq x \leq 2\pi$ on the same set of axis. 1
- (ii) By considering $f(x) = 3 \sin x - x$, show that the curve $y = 3 \sin x$ and the line $y = x$ meet at a point P whose x coordinate is between $x = 2.2$ and $x = 2.4$. 1
- (iii) Using one application of Newton's method, starting at $x = 2.3$, find an approximation for the x coordinate of P . Give your answer to two decimal places. 2
- (b) (i) Find the exact value of $\tan^{-1}(\sqrt{3}) - \tan^{-1}(-1)$. 1
- (ii) Hence or otherwise, find the area bounded by the curve $y = \frac{1}{4 + x^2}$, the x -axis and the ordinates $x = -2$ and $x = 2\sqrt{3}$. 2
- (c) Find the exact value of $\cos\left(\sin^{-1}\left(-\frac{3}{4}\right)\right)$. 2
- (d) (i) Given $P(x) = x^3 + 3x^2 - 10x - 24$, show that $(x + 2)$ is a factor of $P(x)$ and express $P(x)$ as the product of its linear factors. 2
- (ii) Hence solve the inequality $x^3 + 3x^2 - 10x > 24$ 1

End of Question 2

Question 3 (12 marks) Begin a NEW page.

(a) Use the substitution $u = 1 - x$ to find the exact value of $\int_0^1 x\sqrt{1-x} \, dx$. 3

(b) At any point on the curve $y = f(x)$ the gradient function is given by $\frac{dy}{dx} = 2\cos^2 x + 1$. If $y = \pi$ when $x = \pi$, find the value of y when $x = 2\pi$. 4

(c) (i) Use long division to show that 1

$$\frac{x^3 - 2x^2 + 3x - 1}{x^2 + 3} = x - 2 + \frac{5}{x^2 + 3}$$

(ii) Hence find an expression for 2

$$\int \frac{x^3 - 2x^2 + 3x - 1}{x^2 + 3} \, dx$$

(d) Use an appropriate compound angle formula to find the exact value of 2

$$\cos\left(\tan^{-1}\frac{1}{2} + \sin^{-1}\frac{1}{4}\right)$$

End of Question 3

Question 4 (13 marks) Begin a NEW page.

(a) (i) Find $\frac{d}{dx}(x \tan^{-1} x)$. 1

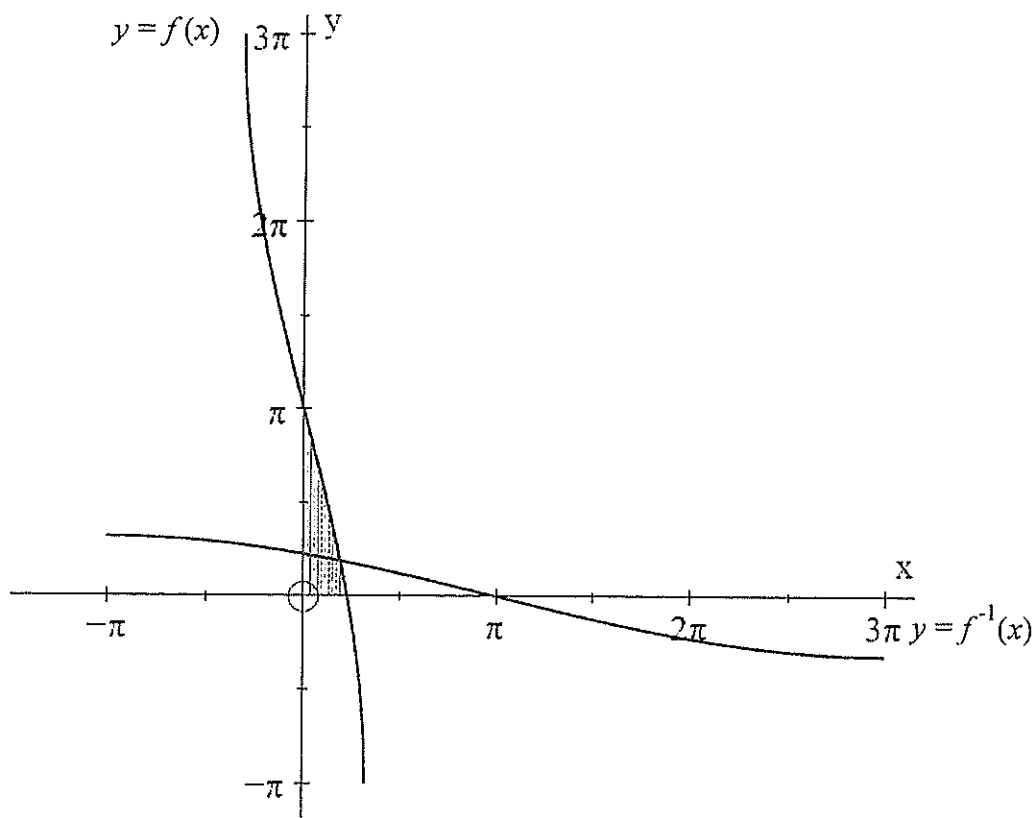
(ii) Hence or otherwise find the exact value of $\int_0^1 \tan^{-1} x \, dx$. 3

(b) Use the substitution $u = e^x$ or otherwise show that 3

$$\int_0^{\ln 10} \frac{3}{1 + 2e^{-x}} \, dx = 6 \ln 2$$

Question 4 continues on the next page

(c)



The graph shows the curves $y = f(x)$ and its inverse $y = f^{-1}(x)$ where $f(x) = \pi - 4\sin^{-1}x$

(i) Find the exact value of x where the curve $y = f(x)$ cuts the x axis. 1

(ii) Find the equation of the inverse function $y = f^{-1}(x)$. 1

(iii) Explain why the area bounded by $y = f(x)$ in the first quadrant is given by 1

$$\text{Area} = \int_0^{\pi} \sin\left(\frac{\pi}{4} - \frac{x}{4}\right) dx$$

(iv) Find the exact value of the area. 2

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

HSC - Extension 1 Task 2 2010 - Solutions

Question 1 (12 marks)

calc
2

com
3

a) $P(x) = 2x^3 - x^2 - 8x + 4$

i) $\alpha + \beta + \gamma = \frac{1}{2}$ ✓

ii) $\alpha\beta + \beta\gamma + \alpha\gamma = -4$ ✓

iii) $\alpha\beta\gamma = -2$ ✓

iv) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ ✓
 $= \left(\frac{1}{2}\right)^2 - 2(-4)$
 $= 8\frac{1}{4}$ ✓

part(iv) caused some problems. Ensure you know this rule or know how to get to it

b) $f(x) = 3\cos^{-1} 2x$ i) $D: -1 \leq 2x \leq 1$

$-\frac{1}{2} \leq x \leq \frac{1}{2}$ ✓

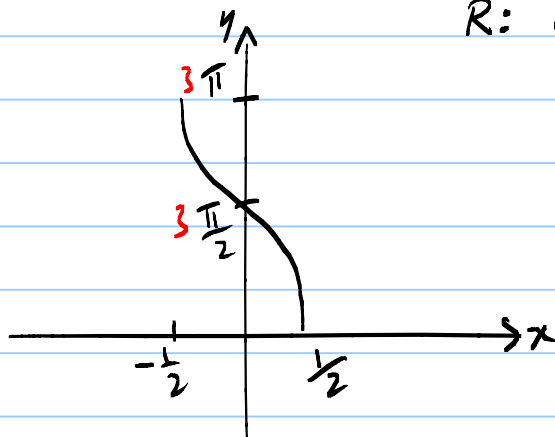
$R: 0 \leq y \leq \pi$

$0 \leq \frac{y}{3} \leq \pi$ ✓

com 3

Very well done!

ii)



c) $\int_0^4 \frac{3}{\sqrt{16-x^2}} dx = 3 \int_0^4 \frac{dx}{\sqrt{4^2-x^2}}$

$= 3 \left[\sin^{-1} \left(\frac{x}{4} \right) \right]_0^4$ ✓

Ca
2

$= 3\frac{\pi}{2}$ ✓

$$d) \quad f(x) = \frac{x+1}{x+2}$$

$$f^{-1}(x) : \quad x = \frac{y+1}{y+2}$$

$$xy + 2x = y + 1 \quad \checkmark$$

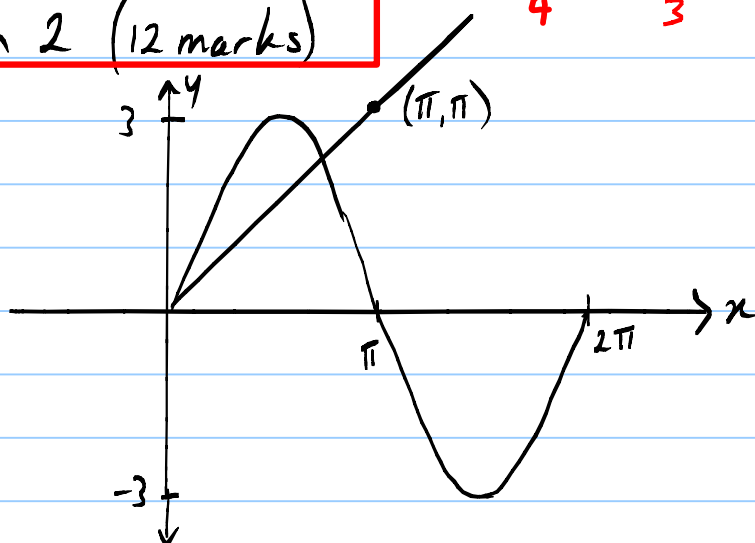
$$xy - y = 1 - 2x$$

$$y(x-1) = 1-2x$$

$$y = \frac{1-2x}{x-1} \quad \checkmark$$

Question 2 (12 marks)

a) i)



Calc 4
Comm 3
reas 2

✓ Comm 1

The location of the line $y=x$ was not well done. Find some points on your calculator $(\frac{\pi}{2}, \frac{\pi}{2})$ (π, π) and plot the location carefully

ii) $f(x) = 3\sin x - x$ $f(2.2) = 0.225$
 $f(2.4) = -0.374$

to find the point of intersection we solve $y = 3\sin x$ & $y = x$ simultaneously, this results in the equation $3\sin x - x = 0$ since $f(2.2) > 0$ and $f(2.4) < 0$ and the curve is continuous, there must be a solution between these values.

Comm 1

Some reasons were too brief. Here are all the details you should include.

iii) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $= 2.3 - \frac{(3\sin 2.3 - 2.3)}{(3\cos 2.3 - 1)}$
 $= 2.28$

Calc 2

⊗ Don't round off too early.
 ⊗ Make sure your calculator is in radians. Write the formula so you don't get mixed up.

$$\begin{aligned}
 \text{b) i) } & \tan^{-1}(\sqrt{3}) - \tan^{-1}(-1) \\
 &= \frac{\pi}{3} + \frac{\pi}{4} \\
 &= \frac{7\pi}{12}
 \end{aligned}$$

Just use your calculator because they are exact ratios. It's too messy using $\tan(A-B)$

$$\begin{aligned}
 \text{ii) } A &= \int_{-2}^{2\sqrt{3}} \frac{1}{4+x^2} dx \\
 &= \left[\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) \right]_{-2}^{2\sqrt{3}} \\
 &= \frac{1}{2} \left(\tan^{-1}\sqrt{3} - \tan^{-1}(-1) \right) \\
 &= \frac{1}{2} \times \frac{7\pi}{12} \\
 &= \frac{7\pi}{24}
 \end{aligned}$$

Calc 2

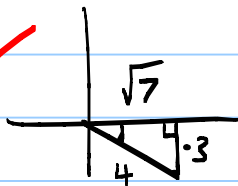
This is a very basic integration directly from the table of standard integrals.

No one should get this wrong.

$$\text{c) } \cos\left(\sin^{-1}\left(-\frac{3}{4}\right)\right)$$

$$\text{let } x = \sin^{-1}\left(-\frac{3}{4}\right) \quad \text{Quadrant 4}$$

$$\sin x = -\frac{3}{4}$$



$$\therefore \cos x = \frac{\sqrt{7}}{4}$$

Reas 2

It is highly recommended to draw the diagram in the correct location clearly showing signs in Quadrant 4.
 $\Rightarrow$ greater success

d) $P(x) = x^3 + 3x^2 - 10x - 24$

$$P(-2) = -8 + 12 + 20 - 24 = 0$$

$\therefore (x+2)$ is a factor ✓

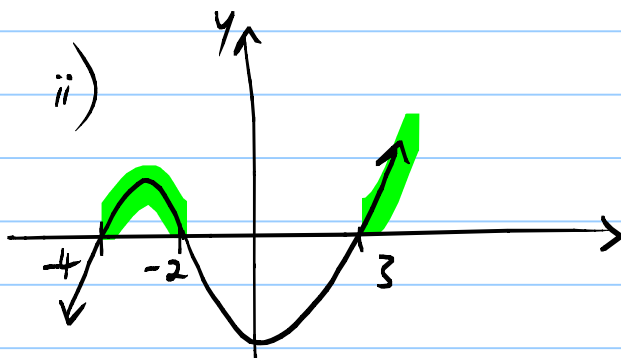
*** make a conclusion**

$$\therefore P(x) = (x+2)(x^2+x-12)$$

$$= (x+2)(x-3)(x+4) \quad \checkmark$$

$$\begin{array}{r} x^2 + x - 12 \\ x+2 \overline{) x^3 + 3x^2 - 10x - 24} \\ \underline{x^3 + 2x^2} \\ x^2 - 10x \\ \underline{x^2 + 2x} \\ -12x - 24 \\ \underline{-12x - 24} \\ 0 \end{array}$$

It was most disappointing to see any students unable to do long division. You must practise this technique. It will very likely appear in both the Trial HSC and HSC.



$$x^3 + 3x^2 - 10x > 24$$

$$x^3 + 3x^2 - 10x - 24 > 0$$

$$-4 < x < 2, x > 3 \quad \checkmark \text{ com } 1$$

This is a very standard basic question. If you couldn't do it please do more practice on Polynomials before The Trial & HSC.

Question 3 (12 marks)

Calc
5

Reas
6

a) $u = 1 - x, x = 1 - u$

$$\frac{du}{dx} = -1$$

$$du = -dx$$

$$x = 0, u = 1$$

$$x = 1, u = 0$$

$$\int_0^1 x \sqrt{1-x} dx$$

$$= -\int_1^0 (1-u) u^{\frac{1}{2}} du$$

$$= -\int_1^0 u^{\frac{1}{2}} - u^{\frac{3}{2}} du$$

$$= \int_0^1 u^{\frac{1}{2}} - u^{\frac{3}{2}} du$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= 0 - \left(\frac{2}{5} - \frac{2}{3} \right)$$

$$= \frac{4}{15}$$

Calc 3

You can also use the fact that $\int_a^b = -\int_b^a$ to change this here $-\int_1^0 = \int_0^1$ to make it easier.

b) $\frac{dy}{dx} = 2\cos^2 x + 1$

$$y = \int 2\cos^2 x + 1 dx$$

$$= \int 2\left(\frac{1}{2}(1 + \cos 2x)\right) + 1 dx$$

$$= \int (1 + \cos 2x + 1) dx$$

$$= \int (\cos 2x + 2) dx$$

$$\begin{aligned} * \cos 2\theta &= 2\cos^2 \theta - 1 \\ 2\cos^2 \theta &= \cos 2\theta + 1 \end{aligned}$$

* You must know the substitutions

$$\begin{aligned} \int \cos^2 x dx &= \int \frac{1}{2}(1 + \cos 2x) dx \\ \int \sin^2 x dx &= \int \frac{1}{2}(1 - \cos 2x) dx \end{aligned}$$

and how to apply them

$$y = \frac{1}{2} \sin 2x + 2x + c \quad \checkmark$$

$$\pi = \frac{1}{2} \sin 2\pi + 2\pi + c$$

$$\pi = 2\pi + c$$

$$c = -\pi \quad \checkmark$$

$$\therefore y = \frac{1}{2} \sin 2x + 2x - \pi$$

at $x = 2\pi$

$$y = \frac{1}{2} \sin 4\pi + 4\pi - \pi$$

$$y = 3\pi \quad \checkmark$$

Evaluate $\sin 2\pi = 0$ to simplify at this stage. Quite a few students made it harder by not doing that step.

(Reas 4)

$$\begin{array}{r} \text{c) i)} \quad x^2+3 \overline{) \begin{array}{r} x^3 - 2x^2 + 3x - 1 \\ x^3 \\ \hline -2x^2 \\ -2x^2 \\ \hline -6 \\ 5 \end{array}} \end{array}$$

Long division again!
You must be able to do it

$$\therefore x^3 - 2x^2 + 3x - 1 = (x^2 + 3)(x - 2) + 5$$

$$\frac{x^3 - 2x^2 + 3x - 1}{x^2 + 3} = x - 2 + \frac{5}{x^2 + 3}$$

$$\text{ii)} \quad \int \frac{x^3 - 2x^2 + 3x - 1}{x^2 + 3} dx$$

$$= \int x - 2 dx + \int \frac{5}{x^2 + 3} dx$$

$$= \frac{x^2}{2} - 2x + 5 \int \frac{1}{x^2 + (\sqrt{3})^2} dx$$

$$= \frac{x^2}{2} - 2x + \frac{5}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + c \quad \checkmark \quad \checkmark$$

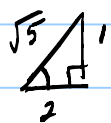
(*) Everyone should have recognised this integration using the table of standard integrals especially in an Inverse Functions test.
(Calc 2)

$$d) \cos \left(\tan^{-1} \left(\frac{1}{2} \right) + \sin^{-1} \left(\frac{1}{4} \right) \right)$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\text{let } \alpha = \tan^{-1} \frac{1}{2}$$

$$\tan \alpha = \frac{1}{2}$$

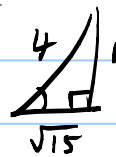


$$\cos \alpha = \frac{2}{\sqrt{5}}$$

$$\sin \alpha = \frac{1}{\sqrt{5}}$$

$$\beta = \sin^{-1} \frac{1}{4}$$

$$\sin \beta = \frac{1}{4}$$



$$\cos \beta = \frac{\sqrt{15}}{4}$$



$$\therefore \cos(\alpha + \beta) = \frac{2}{\sqrt{5}} \times \frac{\sqrt{15}}{4} - \frac{1}{\sqrt{5}} \times \frac{1}{4}$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{4\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= \frac{\sqrt{3}}{2} - \frac{\sqrt{5}}{20}$$

$$= \frac{10\sqrt{3} - \sqrt{5}}{20}$$



Reas 2

This part was pretty well done. Make sure you can rationalise surds.

Question 4 (13 marks)

Calc 5 Com 1 R 4

a) i) $\frac{d}{dx} x \tan^{-1} x$

$u = x \quad v = \tan^{-1} x$
 $u' = 1 \quad v' = \frac{1}{1+x^2}$

$= \tan^{-1} x + \frac{x}{1+x^2}$

Well done!

ii) Hence $\int_0^1 \tan^{-1} x + \frac{x}{1+x^2} dx = x \tan^{-1} x$

R 4

$\int_0^1 \tan^{-1} x dx + \int_0^1 \frac{x}{1+x^2} dx = [x \tan^{-1} x]_0^1$

$\int_0^1 \tan^{-1} x dx = [x \tan^{-1} x]_0^1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$

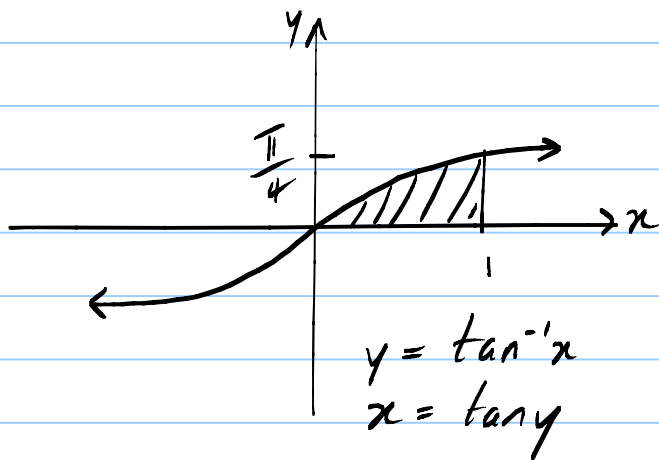
$= [x \tan^{-1} x]_0^1 - \frac{1}{2} [\log_e(1+x^2)]_0^1$

$= \left(\frac{\pi}{4}\right) - \frac{1}{2} (\log_e 2 - \log_e 1)$

$= \frac{\pi}{4} - \log_e \sqrt{2}$

many students were able to determine the initial line of working but then had difficulty determining which part to integrate and with the integration itself. You should recognise that $\int \frac{x}{1+x^2}$ will be a log

alternate method



$$\int_0^1 \tan^{-1} x = \text{rectangle} - \int_0^{\frac{\pi}{4}} \tan y \, dy$$

$$= \frac{\pi}{4} + \int_0^{\frac{\pi}{4}} \frac{-\sin y}{\cos y} \, dy$$

$$= \frac{\pi}{4} + \left[\log_e(\cos y) \right]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} + \left(\log_e \frac{1}{\sqrt{2}} - \log_e 1 \right)$$

$$= \frac{\pi}{4} + \log_e \sqrt{2}^{-1}$$

$$= \frac{\pi}{4} - \log_e \sqrt{2}$$

b) $u = e^x$ $\frac{1}{u} = \frac{1}{e^x}$

$$\frac{du}{dx} = e^x$$

$$dx = \frac{du}{e^x}$$

when $x=0$, $u=1$
 $x=\ln 10$, $u=10$

$$\int_0^{\ln 10} \frac{3}{1+2e^{-x}} \, dx = \int_1^{10} \frac{3}{1+\frac{2}{u}} \times \frac{du}{u}$$

$$= \int_1^{10} \frac{3}{\frac{u+2}{u}} \times \frac{du}{u}$$

$$= 3 \int_1^{10} \frac{u}{u+2} \times \frac{du}{u}$$

$$= 3 \int_1^{10} \frac{1}{u+2} \, du$$

$$= 3 \left[\log_e(u+2) \right]_1^{10}$$

$$= 3 (\log_e 12 - \log_e 3)$$

$$= 3 \log_e 4$$

$$= 3 \log_e 2^2$$

$$= 6 \ln 2$$

The substitution here was a little tricky because e^{-x} was in the denominator and the fractions caused problems for some students. Most know what needs to be done but are getting caught with the algebra

alternate method

$$\begin{aligned} & \int_0^{\ln 10} \frac{3}{1 + \frac{2}{e^x}} dx \\ &= \int_0^{\ln 10} \frac{3}{\frac{e^x + 2}{e^x}} dx \quad \checkmark \\ &= 3 \int_0^{\ln 10} \frac{e^x}{e^x + 2} dx \\ &= 3 \left[\log_e(e^x + 2) \right]_0^{\ln 10} \quad \checkmark \\ &= 3 \left(\log_e(e^{\ln 10} + 2) - \log_e 3 \right) \\ &= 3 \left(\log_e 12 - \log_e 3 \right) \\ &= 3 \log_e 4 = 6 \ln 2 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{c) i) } f(x) &= \pi - 4 \sin^{-1} x \\ 0 &= \pi - 4 \sin^{-1} x \\ 4 \sin^{-1} x &= \pi \\ \sin^{-1} x &= \frac{\pi}{4} \\ x &= \sin \frac{\pi}{4} \\ x &= \frac{1}{\sqrt{2}} \quad \checkmark \end{aligned}$$

parts (i) & (ii) were very well done. Explanations in part (iii) were often too brief you needed to explain 2 things, first why 0 to π and secondly why $\sin(\frac{\pi}{4} - \frac{x}{4})$ gave you the required area

ii) inverse $x = \pi - 4\sin^{-1}y$
 $\sin^{-1}y = \frac{\pi}{4} - \frac{x}{4}$ ✓
 $y = \sin\left(\frac{\pi}{4} - \frac{x}{4}\right)$

iii) The area bounded by $y = f(x)$ in the first quadrant would be given by $\int_0^{\frac{\pi}{4}} \pi - 4\sin^{-1}x \, dx$

but this is not easily calculated.

the area bounded by $f(x)$ in the first quadrant is equal to the area bounded by $f(x)$ and the y-axis from $y=0$ to $y=\pi$

✓
 com (This area is also equal to the area bounded by $f^{-1}(x)$ and the x-axis from $x=0$ to $x=\pi$)

ie Area = $\int_0^{\pi} \sin\left(\frac{\pi}{4} - \frac{x}{4}\right) dx$

iv) Area = $\int_0^{\pi} \sin\left(\frac{\pi}{4} - \frac{x}{4}\right) dx$

= $\left[+4 \cos\left(\frac{\pi}{4} - \frac{x}{4}\right) \right]_0^{\pi}$ ✓

Ca
2

= $4 \left(\cos 0 - \cos \frac{\pi}{4} \right)$

= $4 \left(1 - \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \right)$

= $4 \left(\frac{2 - \sqrt{2}}{2} \right)$

= $4 - 2\sqrt{2}$ ✓

quite good but
 some careless
 errors with
 negative signs